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M 004 VALVES

1 SCOPE OF WORK

The Bidder shall supply all valves required for the project in accordance with Part 4, Subsection 1 .2, “Summary of work” including and but not limited to the following :

- Cast iron and steel gate, globe and angle valves. Gate valves for critical systems shall be of the wedge gate type as supplied by approved manufacturers.
- Check valves
- Butterfly valves
- Ball valves
- Plug valves
- Safety relief valves
- Its associated control and electrical equipment

The requirements set forth in this Specification are the minimum, and are to be supplemented by the Bidder's requirements.

All necessary main, auxiliary, instrumentation valves etc. to assure a proper, complete, safe, functional and maintainable systems shall be furnished, delivered and installed by the Bidder.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.

The codes and standards of the following organizations extracted from Part 4,

Subsection 2.5 “Codes and Standards” or other equivalent international standards, are specifically applicable to the design, manufacture and testing of the equipment included in this Specification.

ASTM	American Society of Testing and Materials
ASME	American Society of Mechanical Engineers

NFPA	National Fire Protection Association
ANSI.	American National Standards Institute, Inc.
DIN	Deutsches Institut für Normung
AWWA	American Water Works Association
IEC	Electro Technical Commission
IEEE.	Institute of Electrical and Electronic Engineers
ISO	International Standards Organization
MSS	Manufacturers Standardization Society of the Valve and Fitting Industry
NEMA	National Electrical Manufacturers Association

3 DESIGN AND OPERATING CONDITIONS

The design basis for the valves shall be selected in accordance with the system requirements where the valve is located. When an isolating valve is located at the junction of two piping systems having different pressure temperature ratings, the subject isolation valve shall meet the requirements of the higher class (most stringent) piping system.

For environmental considerations refer to Part 4, Subsection 2.16

4 DETAILED REQUIREMENTS

4.1 General

All valves shall be suitable for the service conditions under which they are required to operate. The design, construction and choice of materials shall take into account all operational deviations including pressure surge and thermal shock.

All valves shall have platform for operation and maintenance. Necessary handling facility shall be available for each valve.

Valve trim materials for steam services generally should be ASME 316 type stainless steel satellite sealing rings and seats; all other materials should be

dimensionally stable over the operating temperature range. Where the valves include the use of stellite or other hardfacing material to the valve seats, there shall be a differential hardness of not less than 50 and not more than 100 Brinell between fixed and moving seats, with the harder material on the fixed seats. Valves having a back-seating facility are preferred, particularly in critical positions

Unless otherwise specified, the bodies of valves shall be forged or cast steel. Gun metal or cast iron valves shall not be used in any fuel or oil/gas system or any feedwater or similar system where shock loads could result in sudden fracture.

Isolating valves for steam services should normally be of the parallel slide or split wedge gate type.

Gate valves should be such that the bore of the fully open valve is unobstructed by any part of the gate.

Where the blade of a butterfly valve requires significant thickening to provide adequate stiffness a through-flow (low loss) design is preferred.

Valves operating under sub-atmospheric conditions shall be with glands and seating rings water sealed. All necessary sealing water header tanks and piping shall be provided under this Contract. Where temperatures permit, use may be made of diaphragm or "pinch-type" valves which do not close due to atmospheric pressure.

Diaphragm valves shall be used for services carrying chemicals or aggressive fluids, and also for salt water services.

Valves for lubricating oil services shall be flanged, having steel bodies with stainless trim.

Any valve which is designed for uni-directional flow shall have an arrow embossed or cast on the valve body clearly indicating the required flow direction.

All valves, unless otherwise approved or specified, shall be of the external rising spindle type. Where desirable to protect the spindle against ingress of dirt, or where the position of the valve may create a hazard to operators when the spindle is extended, suitable spindle covers shall be provided. The spindles and operating gear of all valves for use outdoors shall have weather and dust proof protection.

Any gear or bevel wheels used to transmit motion shall be of cast steel or approved quality cast iron with machine cut teeth. No thrust from the valve shall be transmitted to the extension spindles and valve pedestals shall not be mounted directly on floor plating. Any floor steel work trimmers for supporting pedestals shall be provided by the Bidder.

Special attention shall be given to the operating mechanism and correct lubrication of all valves to ensure a minimum of maintenance and ease of operation.

All valves shall be positioned so as to be readily accessible for operation and maintenance from permanent floors, galleries or access platforms.

As far as practicable any maintenance or replacement of wearing parts shall be possible with the valves in situation, but for valves with connecting branches of less than 50 mm nominal bore full access to moving components only will be accepted. The seats of all steam valves shall be accessible for machining for purposes of reclamation or repair.

Eyebolts or similar facilities shall be provided, where necessary, to facilitate the handling of heavy valves or components.

No valve shall be installed with its operating spindle gland below the horizontal center line so as to avoid dirt entering the gland and to ensure complete drainage.

When valves are required to be locked in position for operation they shall be provided with a approved locking system on the spindel (no chain) and with a master key system.

All valves shall be provided with labels or nameplates. These labels shall include in the inscription a valve "name", or abbreviated description of the valve duty, and a unique plant reference number for the valve.

Within 2 months of award of Contract, a valve numbering scheme to cover the complete Contract Works shall be agreed. After approval of the numbering scheme, the Bidder shall draw up a valve schedule to cover the Contract Works. The scheduled details of each valve shall include the valve number, its title as it appears on the valve label, the nominal size of the valve, its design pressure and temperature, the Manufacturer's name and model number and a brief description of the valve materials. The valve schedule shall be subject to the Engineer's approval. The finalized valve schedule shall be included in the operating and maintenance manuals, provided under this Contract.

Where a modulating valve in high-pressure service has a full-flow emergency by-pass, the latter shall also be capable of modulating duty.

All high pressure valves of 150 mm bore and above shall be provided with by-passes connected to the valve body. All high pressure gate valves shall be provided with pressure sealed bonnet (breechlock) shall be provided with pressure relief device (safety valve, rupture disc etc.). Maintenance of rupture discs shall be possible during the valve in service.

All valve applications that will be used for isolating purposes at high pressure shall offer double isolation by means of two valves in series.

All valves shall be provided with weld ends unless otherwise specified.

Valves shall be provided in accordance with ANSI B31.1, "Power Piping Code", requirements, or other governing design codes, where applicable. Valves shall comply with standard pressure classes shown in ANSI B16.34.

Material for each valve shall conform to the material requirements referenced in Clause 4.2. herein.

Valves shall be designed for the maximum working pressure of designated pressure class and shall be suitable for the conditions specified.

The design, materials, fabrication, construction, testing and certification of all valves shall conform to the requirements of ANSI B16.34 and ANSI B31.1 of the latest issue.

All valves furnished shall be in accordance with the applicable requirements of the following

Pressure-Temperature Ratings:

Valves shall have pressure-temperature ratings equal to or greater than the ratings set forth in the following ANSI Standards:

- Flanged Valves: ANSI B16.5 (Iron & Steel) and B16.24 (Brass)
- Welding End Valves : ANSI B16.34

Manufacture : In respects other than those above, valves shall be designed and manufactured in accordance with the following

- Butt-welding or Socket Weld End Valves :Either ANSI B 16.5 or ANSI B16.34.
- Flanged-End Valves ANSI B16.5

Socket-Weld Valves: Of adequate design and construction for the specified service conditions. Bidder shall certify, in each case, that these valves conform to the applicable requirements of the governing code.

Special design valves for pressure-temperature conditions exceeding the allowable ratings or Classes set forth in ANSI B16.5 or ANSI B16.34 shall be permanently marked on the valve body with the specified pressure-temperature rating for which Bidder shall certify that the valve is suitable and, in each such case, that the valve conforms to the applicable requirements of the governing codes(s).

The valve size shall be designated by the actual body size.

Face-to-face dimensions of flanged valves and end-to-end dimensions of butt-welding valves shall conform to the applicable requirements of ANSI B16.10.

4.1.1 Weld End Connections

Butt-welding end preparations shall conform to the applicable details as accepted by the Owner. Where special wall pipe is indicated for butt-weld end valve connections Bidder shall verify with the Owner the actual dimensions of the pipe and weld end preparation prior or machining of the ends of such valves if interfacing piping is supplied and erected by Bidder.

Socket-welding end preparations shall conform to the applicable requirements or ANSI B16.11. Where special wall pipe is indicated for socket-weld valve connections the following shall apply:

- Hubs of socket-weld ends shall be machined to suit the O.D. of the special wall pipe, with clearance and tolerance comparable to that set forth in ANSI B1 6.11 for schedule pipe connections.
- Bore of socket-weld ends shall be equal to or greater than the inside diameter of the special wall pipe with tolerances comparable to that set forth in ANSI B16.11 for schedule pipe connections unless otherwise accepted by the Owner.
- The wall thickness of socket weld ends (including hubs), after machining, shall not be less than the allowable minimum set forth for fittings in Article 6 of ANSI B16.11.

Valves shall be full-ported except when a longer-body size is required for structural strength.

Valves and operators shall be designed so that they will function properly when installed in either horizontal, vertical or inclined pipe runs.

The valves shall be designed for a service life of 30 years considering normal maintenance, accounting for corrosion and material life.

Valves shall visually indicate the direction of flow and the open and closed positions.

Valve design shall be such that the stem shall not be capable of removal from the valve while the valve is under pressure by the removal of the stem seal retainer alone.

All valves shall be capable of withstanding the hydrostatic pressure test conducted at 1.5 times the design pressure at ambient temperatures.

All valves shall provide tight shut off and shall be furnished with replaceable seats.

All valves shall function smoothly without sticking, rubbing, vibrating or scoring on opening and closing.

4.1.2 Cast Iron Valves

Cast iron gate and swing check valves in sizes 2 in. and larger shall conform to the requirements of

MSS SP-70 - Cast Iron Gate Valves, Flanged and Threaded Ends.

MSS SP-71 - Cast Iron Swing Check Valves, Flanged and Threaded Ends

MSS SP-85 - Cast Iron Globe and Angle Valves, Flanged and Threaded Ends

Other types and sizes of cast iron valves shall also conform to the applicable requirements of the above MSS Standards and shall be acceptable to the Owners. Pressure-temperature ratings of all cast iron valves shall be in conformance with those set forth in Table 1 of MSS SP-70, MSS SP-71 and MSS-SP-85, unless otherwise indicated.

4.2 Materials

Materials for each valve shall be suitable for the intended service and in accordance with the governing design codes.

Material for steel valves shall be of the type indicated and shall be in conformance with the requirements for use at the specified service conditions.

Bolting and nuts shall be of materials most suitable of the specified service conditions. Bolting and nuts for steel valves shall be in conformance with ASTM A193 and A194 respectively.

Materials not definitely specified herein shall be of a type most suitable for the specified service conditions in each case and shall conform to the applicable code requirements. Material shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product.

Construction materials shall be selected to avoid use of alloys containing low melting point elements, including (but not limited to) lead, mercury, sulphur etc. Copper alloys shall not be allowable to come in contact with stainless steel.

Nitriding treatments or chrome plating on any surface exposed to the working fluid are prohibited.

Care must be taken to avoid use of any welding flux, gaskets, antitrust compound or other chemical employed in the fabrication or production that contain chlorides or other constituents capable of inducing stress corrosion in stainless steel or copper alloys.

All stainless steel components shall be handled using slings, ropes or other media free from ferrous contamination. All grinding performed on stainless steel components shall be accomplished using iron free compounds, files or burring tools which have not been used previously on iron or carbon steel.

For the purpose of conformance with the requirements of materials, design, manufacture and examination, the pressure boundary components of the valve shall be defined as the valve body, bonnet, body-bonnet bolting, valve disc and related pressure retaining welds.

4.3 Gate, Globe and Angle Valves

4.3.1 Bonnets

Valve bonnets shall be of the outside screw and yoke (OS & Y) design with rising stem unless otherwise indicated. Bonnet to body connections shall conform to the following

a) 80 mm size and larger

Pressure Class 125 Cast iron.....Bolted
Pressure Classes 150 and 300 steel..... Bolted
Pressure Classes 600 through 2500 and special design steel ..Pressure seal

b) 50 mm size and smaller

Pressure Class 600 forged steel..... Bolted or welded

Pressure Classes 1500 through 2500 and special design forged steel.....
welded, integral or pressure seal.

Stuffing boxes shall be of adequate depth and dimensions to assure tightness and long packing life. Stuffing box shall have bolted packing gland and shall be provided with lantern-type gland or condensate chamber where required. Lantern rings shall be drilled with tapped holes to facilitate ring removal.

Valve shall have back seats to permit repacking of the shaft seals under full pressure when in wide open position, and to protect the packing in service. Valves shall be of a design which permits adjustment, removal and replacement of the seats, discs, shafts, bearings and seals in the field.

The valve packing shall be guaranteed but Bidder to be suitable, in each case, for the specified service conditions.

4.3.2 Seats and Discs

Cast iron valves (80 mm and larger) shall have bronze removable seat rings and bronze or bronze faced disks or wedges.

Steel valves (80 mm and larger)

- a) Gate valves shall have flexible wedges discs, except valves for HP steam and HP feedwater systems shall have parallel type discs.
- b) Globe and angle valves shall have swivel or integral stem disks.
- c) Gate, globe and angle valves in Pressure Class 600 and higher shall have Stellite seating.
- d) Globe and angle valves in Pressure Class 600 and higher shall have seat facing materials integrally welded to the valve body.
- e) Valves in Pressure Class 300 and less shall have renewable or welded-in seat rings.

4.3.2.1 Steel valves (50 mm and smaller)

- a) Globe and angle valves shall have swivel discs.
- b) Valves shall have stiletted seating surfaces integrally welded to the valve components.

4.3.3 Trim Materials

All trim materials shall be suitable for the specified service conditions.

4.3.4 Stems

Steel valves shall have alloy steel stems. Cast iron valves shall have forged alloy steel stems for steam service and bronze stems for water service.

4.4 Check Valves

4.4.1 Type

Check valves shall be of the non-slam type on pump discharges while they may be swing, piston disk or lift type, in other applications.

4.4.2 Cover

Valve cover to body connections shall conform to the following 80 mm size and larger

Pressure Class 125 Cast iron.....Bolted

Pressure Class 300 and less steel.....Bolted

Pressure Classes 600 through 2500 and special design steel..... Pressure seal

50 mm size and smaller :

Pressure Class 600 forged steel.....Bolted

Pressure Classes 1500 through M 2500 and special design steel.....Welded or pressure seal

4.4.3 Seating Surfaces

Cast iron valves 80 mm and larger:

- Valves shall have bronze removable seating rings.
- Valves shall have bronze or bronze faced discs.

Steel valves 80 mm and larger

- Valves in Pressure Classes 600 through 2500 and special design shall have Stiletted seating surfaces.
- Seating surfaces may be integrally welded to the valve body.
- Pressure Class 300 and less valve shall have removable seat rings with hard faced seating surfaces.

Steel valves 50 mm and smaller

- Valves shall have stelitted seating surfaces integrally welded to the valve

- components.

All- trim materials shall be suitable for the specified service conditions.

The valve design in each case shall permit the valve to function properly if installed in horizontal, vertical or inclined pipelines.

4.5 Drains and Bypasses

4.5.1 Drains

Valves used for steam service shall be provided with drain openings in the body. Bonnets on bolted bonnet steam valves shall be provided with drain openings.

Drain outlets on Pressure Class 600 valves and higher shall be provided with pipe nipples socket-welded into the valve body. Drain nipples shall be a minimum of Schedule 80 seamless steel pipe and heavier, if required. And if used for operating temperatures above 412 °C shall be of suitable alloy steel.

4.5.2 Bypasses

Valves for steam and boiler feed service shall be equipped with standard size bypass where pressure equilisation is important prior to opening of the main valve

Each bypass shall be of comparable materials and have two (2) steel globe valves with weld end connections, suitable for the conditions. Bypass valves shall conform to the applicable requirements hereinbefore set forth for main valves.

4.5.3 Drain and Bypass Connections

The location, designation, drilling and preparation of bosses and the size and location of drains and bypasses shall be in conformance with MSS-SP-45 as a minimum standard.

4.5.4 Materials

Piping materials used in bypasses shall be of the same nominal chemical and physical properties as the material of the valve body to which they attach as follows (ASTM A or ASME SA-):

For temperatures up to 413°C

- Pipe106Gr.B or C

- Fittings.....105
- Valves105 or 216- Cr. WCB

For temperatures of 414°C to 512°C

- Pipe.....Gr.P11
- Fittings.....182 Cr. Fl 1 or A217 Gr. WC6
- Valves.....182 Cr. Fl 1 or A217 Gr. WC6

4.6 Butterfly Valves

Butterfly valves shall have bodies of either the flanged or in case of non stream service of lugged wafer type with down stream dismantling. The actuator/manual gearbox shall be mounted direct to the valve body, other than where both actuator and body are fitted with feet for pedestal mounting; no actuator loads shall be transmitted via the pipework. The valve should be mounted with spindle set horizontal. If the valve is designed for unidirectional flow, an arrow shall be cast or embossed in a prominent position on the valve body to indicate that direction.

The valve must be capable of positive shut-off in both directions against full system pressure. Unless an alternative type is proposed by the Bidder and approved by the Owner as being fully suitable for the particular duty, the seating of the disc shall be positive (that is, the disc must not be capable of passing through the shut position); a stop fitted on the actuator/operating lever is not adequate for this purpose.

Unless the Bidder can demonstrate that some alternative design is capable of adequate long-term sealing, the design shall be such that the shaft is offset from the seal ring (that is there shall be an unbroken peripheral seal when the valve is shut).

If the valve body is rubber (or elastomer) lined, the lining should be made by a self containing and replaceable inlet, unless otherwise approved by the Owner.

Internal shaft bearings must be capable of operating dry, that is, not dependent on the presence of the pumped fluid.

4.6.1 High Performance Butterfly Valves

High Performance Butterfly Valves are intended for use in main- cycle power plant systems where high reliability and long life are required. Application Group I valves include valves which can accommodate up to 204°C and pressures up to the full ANSI pressure ratings of Standard Class 1 50 and Standard Class 300 as specified in ANSI 816.34. High performance as indicated in this specification, requires eccentric disc butterfly valves with bi- directional shutoff capability and replaceable valve seats.

Materials shall be suitable for the required design conditions or the service conditions specified in Part 4, Technical Specification, M-001 Piping, and shall conform with the requirements of MSS SP - 67 or AWWA C 504 except as specified herein. In all cases the disks and stems shall be stainless steel.

The replaceable valve seat shall be constructed of Teflon or 316 Stainless Steel. The valve seat shall be replaceable without requiring the removal of the valve disk or shaft.

Valves shall be provided with a stuffing box and pull-down packing gland. Stuffing boxes shall have sufficient depth to accept at least four rings of packing. The design of the valve and stuffing box assembly shall be such that the packing can be adjusted or replaced without removing the valve from the line or removing the valve shaft. Valves in sizes 600 mm and larger shall be provided with means of replacing the packing without removal of the valve operator. Packing material shall Grafoil, or Teflon as required by the service, or an alternative acceptable to the Owner.

All valves shall receive a hydrostatic shell test at the hydrostatic pressures specified in ANSI B16.34.

All valves shall receive a seat leakage test in accordance with the requirements specified in ANSI B 16.34. Valve seats shall be bubble-tight.

Alternative leakage and hydrostatic test procedures are acceptable if specified by the Bidder in his Bid Proposal and acceptable to the Owner.

All valves shall receive a performance test in accordance with AWWA C504.

4.6.2 Low Performance Butterfly Valves

Low performance butterfly valves are intended for use in low temperature, low pressure main power plant systems where high reliability and long life are re-

quired. Application Group II valves include valves which accommodate up to approximately 65°C and differential pressures across the valve of up to 10 bar.

Valves in sizes 600 mm and smaller shall be bubble-tight at pressures up to the Class 1 SOB rating as specified in AWWA C504, and used at temperatures up to 51°C. Valves in sizes larger than 600 mm shall be leak-tight at pressures specified in Part 4, Technical Specification, M-001 Piping, or higher if required by Bidder's design.

Valve bodies shall be cast iron, heat-treated ductile cast iron in accordance with AWWA C504 or fabricated steel conforming to ASTM A36 or ASTM A 515, Gr. 60. Valve shafts shall be stainless steel in accordance with AWWA C504. (4083 Valve disks shall be cast iron heat-treated ductile cast iron or cast steel in accordance with AWWA C504 or fabricated steel conforming to ASTM A36 or ASTM A515, Gr. 60.

Valves shall be rubber-sealed. Valve seat design shall be in accordance with AWWA C504. In all cases, a corrosion-resistant seating surface of stainless steel or model on the disk edge or body seat shall be provided.

Shaft seals shall be designed for either self-adjusting split-V packing or compression packing with a stuffing box and packing gland. Both designs shall be in accordance with AWWA C 504. Valves in sizes 600 mm and larger shall be provided with means of replacing the packing without removal of the valve operator.

4.6.3 Valve Testing

All valves shall be hydro tested and seat leak tested as per ANSI B16.34 requirements in accordance with pressure listed in the standard.

4.7 Commercial Grade Butterfly Valves

Commercial Grade Butterfly Valves shall not be used.

4.8 Ball Valves

All steel valves shall conform to the requirements of ANSI B 16.5 or ANSI B16.34 of latest issue, whichever is applicable, and all brass valves shall conform to the requirements and limits set forth in ANSI B 16.24, class 150, and shall be of the primary service pressure rating as required by the Specification.

All valves shall be non-lubricated, isomer-sleeved for bi-directional flow.

Valves shall be equipped with manual operators and operating handles, life-time lubricated. These operators shall be of a type that will hold valve in position selected and prevent movement of valve ball or plug by pipeline vibration, pressure inside valve, pressure surges etc.

All ball valves shall be full ported TFE-seated, with stainless steel ball.

4.9 Plug Valves

Plug valves shall be of the tapered plug type polymer or isomer seated.

System Balancing Valves. In addition to the requirements for plug valves, system balancing plug valves shall:

Have a calibrated plug orifice.

Have a 20mm NPT connection on both the upstream and downstream ends of the plug to which Owner may attach pressure gauges.

TFE-seats shall not be used on any valve used for throttling. The Bidder shall propose materials for Owner's review.

4.10 Pressure Relief Valves

Pressure relief valves shall be of the spring-loaded type unless otherwise specified, or unless the owner has given written permission for the use of a different type of valve actuation.

Spring-loaded valves shall be fitted with easing gear which shall not be extended.

Where necessary relief valves shall be supplied with vent pipes, silencers and drains, and with gagging equipment for hydraulic testing following erection and in the long-term maintenance of the plant.

All valves shall be rated for the primary service pressure rating called for in this specification. Safety-Relief Valve Data Sheets or a higher service pressure rating if required.

A valve data sheet per Exhibit 4 shall be submitted by the Bidder for each safety relief valve.

Valves shall be suitable for operating conditions as listed below:

Under variable and constant back pressures as required .

For the design pressures and temperatures required.

Valves shall be sized and designed to accommodate any flashing that may occur across valves upon actuation.

Valves shall be sized and designed for smooth operation at the set pressure without valve chatter.

4.11 Valve Operators

4.11.1 General

All valves shall be provided with manual operators. Manual operators on the 100 mm and larger valves shall be factory mounted with totally enclosed gears and shall be lifetime lubricated. These operators shall be of a type that will hold the flow passage in the position selected and prevent movement of valve plug by pipeline vibration, pressure inside valve, pressure surges, etc. The maximum force to produce the required torque shall be no more than 15 kg on the operating handwheel on the rim of the suitable handwheel. Manual operators shall be equipped with a mechanical limit stop to prevent over travel.

Valves and operators shall be so constructed that valve disc will remain in any set position, full open to full close. Valves may at times be used to throttle the flow. Valves shall be constructed so that in the closed positions, the shaft and disc shall not rotate upon removal or disconnection of the operating mechanism.

Valves shall be furnished with hand wheels, unless other type of actuating device required makes the addition of a hand wheel impractical.

Hand wheels shall be constructed of steel or malleable iron. Steel hammer-blow hand wheels shall be provided, where required for the service conditions. Hand wheels shall be securely fastened to prevent their loosening or falling off due to vibration or other severe operating conditions. Hand wheels shall include arrows indicating opening and closing directions.

When required, valves shall be provided with floor stands for bolting down, suitable for the specified application. Floor stands shall be for hand wheel operation unless required for gear operator, motor operator, or cylinder operator. Extension stems between valve and floor stands shall be furnished. They shall be steel and shall be connected to the valve stem by a sleeve coupling or universal joint, depending upon arrangement.

Where gallery access is impractical valves shall be provided with chain operators as required for accessibility only after review by the Owner.

4.11.2 Pneumatic Operators

Pneumatic operators where specified or required for ease of operation or maintenance shall conform to the following requirements:

A data sheet per Exhibit 2 shall be submitted by the Bidder for each air operated valve.

Each pneumatic valve operator shall be provided with a locally stored energy source (spring or pressure accumulator) for fail- close, fail-open, or fail status quo operation as required for fail-safe operation.

Operators shall be provided with orificing and/or cushioning to control deceleration of each end of stroke.

Pneumatic operators and yokes shall be of steel construction.

Unless otherwise indicated on the data sheets, valve closing and opening travel speed shall be Manufacturer's standard.

Loss of power or of station air pressure shall result in fail-safe position.

Pneumatic operators shall be provided with solenoid control valves with minimum 8 mm NPT or equivalent connections and 6mm or equivalent internal orifice. Solenoid coils shall meet the requirements specified in Part 4, Technical Electrical Specifications, Pneumatic operators shall be provided with a method of adjusting the stem speed acceptable to the Owner. Bidder shall supply piping between solenoid controlled valves and the air operator for valves with locally or remote mounted operators.

Solenoid valves shall be furnished to control air-operated valves. Solenoid valves shall have IP65 (NEMA 4) enclosures for indoor and IPW66 (NEMA 4) for outdoor locations.

Air-operated valves shall have one limit switch which shall operate when valve is in full open position and one limit switch which shall operate when valve is in full closed position. Limit switches shall have at least two normally open and two normally closed electrically independent contacts. Limit switches shall have IP65 enclosures for indoor and IPW66 for outdoor locations.

4.11.3 Motor Operators

Valves shall be provided with motor operators as shown in the P & IDs and/or if required by the Bidder and shall conform to the requirements specified herein and E-210, Technical Specification, Low Voltage Motors.

Valve closing speed shall be 0.10 m/min for motor-operated globe and angle valves, 0.3 m/min for motor-operated gate valves and as required for butterfly valves to eliminate water hammer and pump run out problems unless otherwise noted on the Valve Data Sheets.

4.12 Cleaning and Painting

Valves shall be cleaned and painted in accordance with manufacturer's standard practice and as follows:

Interior of all valve bodies shall be thoroughly blast cleaned using metallic grit. Grit used for stainless steel valves shall be new alumina. All grit blasting materials shall be removed prior to painting.

No paint having an asphaltic base shall be used.

Paint for ferrous metal valves with hot service conditions shall be heat resistant.

Stainless steel valves shall not be painted.

Machined surfaces shall not be painted. Welding end preparations of ferrous metal valves shall be coated with deoxialuminate, or equal, acceptable to the Owner, back to a ring 150 mm from end.

Flange facings, bolt holes and other external machined surfaces of ferrous metal materials shall be coated with a suitable antitrust compound.

4.13 Protection

Materials shall be provided with protection against damage, corrosion and internal contamination during transit and storage, and shall be packaged, packed and prepared for shipment in a manner which will ensure arrival at destination in satisfactory condition. Protection shall include the following:

All openings in equipment furnished shall be securely plugged, capped or otherwise blanked off, sealed with tape and suitably protected against damage and entry of foreign materials and moisture.

Weld ends shall be capped and sealed with suitable, firmly attached protectors. Protectors for beveled ends designed for backing ring or consumable insert shall have a plywood or hard-board liner disc held securely against the beveled end.

Protectors for other weld ends may be metal caps without liner discs.

Protectors for socket weld ends may be plugs.

All flanged connections shall be provided with suitable full-face flanged protectors bolted in place and sealed.

Threaded connections shall be provided with plugs.

4.14 Valve Identification and Marking

4.14.1 Marking

For butt-weld and flanged end steel valves: Per ANSI B16.34.

For cast iron valves: Per MSS SP-70, MSS SP-7 1 and MSS SP-85 as applicable.

For valves not covered above : Per MSS SP-25.

4.14.2 Identification

In addition to the marking above each valve and separate component shall be equipped with a stainless steel metal tag permanently attached and bearing the corresponding identification number and its particular service, as specified in Part 4, Subsection 2.8.

4.15 Valve List

A "Valve List" shall be prepared by the Bidder in accordance with the format approved by the Owner. The Valve List designates information required for design and fabrication.

The Valve List shall have 17 fields of information. Information in the field columns is as follows:

Field 1 : Number

Field 2 : Valve Number

Field 3 : Service Name

Field 4 : P & ID Diagram

Field 5 : Size Diagram

Field 6 : Type

Field 7 : Valve Class

Field 8 : Connection Type

Field 9 : Pipe Class

Field 10 : Operator

a. Type

b. KW

Field 11 : Operating Condition

a. Pressure (bar)

b. Temperature (°C)

Field 12 Design Condition

a. Pressure (bar)

b. Temperature (°C)

Field 13 : Drawing Number

Field 14 : Manufacturer

a. Name

b. Model Number

Field 15 : Remark

Field 16 :Revision Number.

Field 17 :Valve Data Sheet - Reference

4.16 Valve Data Sheets

A valve data sheet shall be prepared by Bidder for all valves in accordance with the data sheets indicated below.

Valves shall conform to all requirements indicated on the valve data sheets.

Valve Operating Conditions: Each valve listed on the Valve List shall be suitable for conditions listed below:

For the maximum normal and design pressures and temperatures listed in the valve data sheet, as required by the Bidders design conditions.

For the fluid contained in the valve as specified in the valve data sheet or specified herein or required by Bidder.

Data Sheets for the following valve types shall be prepared by the Bidder as follows:

Valve type	Typical Data Sheet Included as
Valve List	Exhibit 1
Air Operated	Exhibit 2
Motor Operated	Exhibit 3
Safety Relief	Exhibit 4
Special Manual	Exhibit 5

The valve data sheet will be referenced in column "17" of the Valve List. Manual valves with special accessories will also have valve data sheets.

5 INSTRUMENTATION AND CONTROL

Refer to Part 4. Technical Specification I-001, Instrumentation and Control.

6 ELECTRICAL SPECIFICATIONS

6.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for Valves. Those equipment shall include but not be limited to the following:

- a. Motors including wiring and junction boxes.
- b. All electrical control, monitoring and instrumentation devices and other appurtenances required for this Specification, shall be provided by the Bidder.

For general electrical requirements, see Part 4, E-001 General Electrical Specification.

6.2 Motors

For detailed electrical requirements, see Part 4, Technical Specification E-210 Low Voltage Motors.

EXHIBIT 1 VALVE LIST

NO.	VALVES NUMBER	SERVICE NAME	P&I DIAGRAM	SIZE DIAGRAM	TYPE	VALVE CLASS	CONNECTION TYPE	PIPE CLASS	OPERATOR		OPERATING CONDITION		DESIGN CONDITION		DRAWING NUMBER	MANUFACTURER		REMARKS	REVISION NUMBER	VALVE DATA SHEET REFERENCE
									TYPE	KW	PRESSURE bar (g)	TEMP. T(°C)	PRESSURE bar (g)	TEMP. T(°C)		NAME	MODEL NUMBER			

EXHIBIT 2 AIR OPERATED VALVE

	1. SPECIFICATION NUMBER :	6.	DESCRIPTION & TAG NO.		
	2. PIPE CLASS: SEISMIC CLASS:				
	3. P & ID : M..... SHT.....				
SERVICE	4. PIPE DESIGN TABLE				
	5. DATA TABLE : NONE				
	7. FLUID: RAW WATER OIL STEAM DEMIN WATER AIR OTHER				
		MAXIMUM UNDER NORMAL CONDITION	DESIGN		
	8. INLET PRESSURE bar (g)				
	9. TEMPERATURE (°C)				
	10. MAXIMUM ΔP VALVE MUST OPEN AGAINST bar (g)				
	11. MAXIMUM ΔP VALVE MUST CLOSE AGAINST bar (g)				
	12. FLUID FLOW RATE : (L/SEC)				
	13. ΔP WHEN VALVE FULL OPEN : bar (g)				
	14. BODY SIZE : (MM)				
	BODY	15. BODY TYPE GATE GLOBE DIAPHRAGM BUTTERFLY OTHER ...			
		16. FLOW ACTION OPEN CLOSE BI-DIRECTIONAL			
		17. STEAM LEAKOFF REQUIRED YES NO			
OPERATOR	18. AIR FALLS, VALVES OPEN CLOSE				
	19. ELEC. POWER FAILLS. VALVE OPEN CLOSE				
	20. VALVE CLOSING TIME : AS REQUIRED PER SPECIFIED STEAM SPEED OTHER, SEC.....				
	21. VALVE OPENING TIME : AS REQUIRED PER SPECIFIED STEAM SPEED OTHER, SEC.....				
ACCESS	22. FREQUENCY OF OPERATION CYCLES/YEAR				
	23. POSITION TRANSMITTER YES NO				
	24. POSITIONER YES NO BY VENDOR				
	CONTROLLER NONE PRESSURE TEMPERATURE				
	25. BY OTHERS LEVEL (SPECIAL (SEE REMARKS))				
	26. LOCATION VALVE YOKE REMOTE				
	27. LIMIT SWITCHES NONE FULL OPEN FULL CLOSED				
	28. SELONOID VALVE NONE AC DC SPECIAL (SEE REMARKS) BY OYHERS				
	29. SELONOID VALVE LOCATION VALVE YOKE REMOTE				
	30. AIR SET (W/GAUGES): REQUIRED FOR EACH AIR SUPPLY RANGE				
	31. LOCATION : VALVE YOKE REMOTE				
	32. AIR PRESSURE : RANGETO bar (g) NORMAL.....bar (g)				
REMARKS					
				AIR OPERATED VALVE DATA SHEET	
				MUARA TAWAR	
				COMBINE CYCLE POWER PLANT PROJECT	
DESIGNED BY DATE :		CLIENT : PLN			
CHECKED BY DATE :			JOB NO		
REVISED BY DATE :			DATA SHEET		

EXHIBIT 3 MOTOR OPERATED VALVE

	1. SPECIFICATION NUMBER :	6.	DESCRIPTION & TAG NO.	
	2. PIPE CLASS:	SEISMIC CLASS:		
	3. P & ID : M..... SHT.....			
SERVICE	4. PIPE DESIGN TABLE			
	5. DATA TABLE :	NONE		
	7. FLUID:	RAW WATER	OIL	STEAM
		DEMIN WATER	AIR	OTHER
		MAXIMUM UNDER NORMAL CONDITION		DESIGN
	8. INLET PRESSURE	bar (g)		
	9. TEMPERATURE	(°C)		
	10. MAXIMUM ΔP VALVE MUST OPEN AGAINST	bar (g)		
	11. MAXIMUM ΔP VALVE MUST CLOSE AGAINST	bar (g)		
	12. FLUID VELOCITY@ VALVE INLET	(M/Sec)		
13. ΔP WHEN VALVE FULL OPEN	bar (g)			
BODY	14. BODY SIZE (mm)			
	15. BODYTYPE	GATE	GLOBE	DIAPHRAGM BUTTERFLY OTHER
	16. BODY MATERIAL :	CARBON STEEL : ASTM%CR-MOLY CAST IRON : ASTM.....BRONZE : ASTM 304 SS : ASTM 316: ASTM OTHER		
	17. TYPE AND CONNECTION	INLET		OUTLET
		BUT WELD	FLANGED	BUT WELD FLANGED
		SOC WELD	SCDWED	SOC, WELD SCR WED
		FLANGELESS		FLANGELESS
	18. CONNECTING PIPE SIZE/SCHEDULE	/.....		/.....
	19. WELD AND STANDARD (BUTT WELD ONLY)			
	20. FLOW ACTION	OPEN	CLOSED	BI-DIRECTIONAL NEUTRAL
ACCESS OPR	21. STEAM LEAKOFF REQUIRED:	YES	NO	
	22. MOTOR RATING :	AC	DC	
	23. MOTOR VOLTAGE :	PER PROJECT SPEC. ORVOLTS		
	24. Class IE:	YES	NO	
	25. VALVE CLOSING TIME :	AS REQUIRED FOR SPECIFIED STEAMSPEED OTHER.SEC.....		
	26. VALVE OPENING TIME :	AS REQUIRED FOR SPECIFIED STEAMSPEED OTHER.SEC.....		
	27. POSITION TRANSMITTER	YES	NO	
REMARKS				
		MOTOR OPERATED VALVE DATA SHEET		
		MUARA TAWAR		
		COMBINE CYCLE POWER PLANT PROJECT		
DESIGNED BY	DATE :	CLIENT : PLN		
CHECKED BY	DATE :		JOB NO	
REVISED BY	DATE :		DATA SHEET NO.	

EXHIBIT 4 SAFETY RELIEF VALVE

	1. SPECIFICATION NUMBER :	6. DESCRIPTION & TAG NO.	
	2. PIPE CLASS: SEISMIC CLASS:		
	3. P & ID : M..... SHT.....		
SERVICE	4. PIPE DESIGN TABLE		
	5. DATA TABLE :.....		
	7. FLUID: RAW WATER OIL STEAM DEMIN WATER AIR OTHER		
	8. SET PRESSURE bar (g) NORMAL OPERATING PRESSURE		
	9. TEMPERATURE (°C) DEGREES SUPERHEAT		
	10. BACK PRESSURE bar (g)		
	11. CAPACITY REQUIRED : KG/HR L/SEC		
	12. ACCUMULATION 3% 10% 25% OTHER		
	13. BODY SIZE : (MM)		
	14. GOVERNING CODE SEC. VIII SEC. III SEC. I (BOILER CODE) OTHER (SET REMARKS)		
	15. BODY MATERIAL: CARBON STEEL : ASTM%CR-MOLY CAST IRON : ASTM BRONZE: ASTM..... 304 SS 316 SS : ASTM..... OTHER		
	BODY	16. TYPE AND CONNECTION	INLET OUTLET BUT WELD FLANGED BUT WELD FLANGED SOC WELD SCR WED SOC WELD SCR WED
		17. CONNECTING PIPE SIZE/SCHEDULE	
18. WELD AND STANDARD (BUTT WELD ONLY)		OTHER..... OTHER.....	
19. TEST GAG : YES NO			
ACCESS	20. LIFTING LEVER : NONE UNPACKED PACKED		
	21. BACKPRESSURE BELLOWS YES NO		
REMARKS			
SAFETY RELIEF VALVE DATA SHEET MUARA TAWAR COMBINE CYCLE POWER PLANT PROJECT CLIENT : PLN			
DESIGNED BY	DATE :		
CHECKED BY	DATE :	JOB NO	
REVISED BY	DATE :	DATA SHEET	

EXHIBIT 5 SPECIAL MANUAL VALVE

	1. SPECIFICATION NUMBER :	6. DESCRIPTION & TAG NO.
	2. PIPE CLASS: SEISMIC CLASS:	
	3. P & ID : SHT.....	
SERVICE	4. PIPE DESIGN TABLE	
	5. DATA TABLE : NONE	
	7. FLUID: RAW WATER OIL STEAM DEMIN WATER AIR OTHER	
		MAXIMUM UNDER NORMAL CONDITION DESIGN
	8. INLET PRESSURE bar (g)	
	9. TEMPERATURE (°C)	
	10. MAXIMUM ΔP bar (g)	
	11. FLUID VELOCITY & VALVE INLET (M/Sec)	
	12. ΔP WHEN VALVE FULL OPEN bar (g)	
	BODY	13. BODY SIZE (mm).....
14. BODY TYPE GATE GLOBE DIAPHRAGM BUTTERFLY OTHER		
15. BODY MATERIAL: CARBON STEEL : ASTM%CR-MOLY CAST IRON : ASTM BRONZE: ASTM..... 304 SS 316 SS : OTHER		
16. TYPE AND CONNECTION		INLET BUT WELD FLANGED SOC WELD SCR WED FLANGLESS OUTLET BUT WELD FLANGED SOC WELD SCR WED
17. CONNECTING PIPE SIZE/SCHEDULE		
18. WELD AND STANDARD (BUTT WELD ONLY)		OTHER..... OTHER
19. FLOW ACTION OPEN CLOSED BI-DIRECTIONAL NEUTRAL		
20. STEM LEAKOFF REQUIRED : YES NO		
21. LOCKING DEVICE REQUIRED: NO YES		
22. LIMIT SWITCHES REEQUIRED YES NO FULL OPEN FULL CLOSED		
REMARKS		
		SPECIAL MANUAL VALVE DATA SHEET
		MUARA TAWAR
		COMBINE CYCLE POWER PLANT PROJECT
DESIGNED BY	DATE :	CLIENT : PLN
CHECKED BY	DATE :	JOB NO
REVISED BY	DATE :	DATA SHEET

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